Integrated Academy Of Management And Technology		
Data Structure Algorithm(BCA 3rd Semester)		
BCA-302		
Lesson Plan		
Class	BCA SEM III	
Section	-	
Course Name	Data Structure Algorithm	
Course Code	BCA	
Faculty	Mr. Kamal Nain	
Lecture No.	Unit No	Topic
	Unit I	**Introduction to Data Structures & Arrays**
L-1	1	Introduction to Data Structures - Concept, Need, Characteristics
L-2	2	Types of Data Structures - Primitive & Non-Primitive
L-3	3	Array Fundamentals - Definition, Advantages, Disadvantages
L-4	4	Representation of Single-Dimensional Arrays
L-5	5	Representation of Multidimensional Arrays
L-6	6	Sparse Arrays - Concept, Storage Techniques
L-7	7	Special Matrices - Lower and Upper Triangular Matrices
L-8	8	Tridiagonal Matrices & Vector Representation
	Unit II	**Stacks & Queues**
L-9	9	Introduction to Stacks - Definition, Primitive Operations
L-10	10	Applications of Stacks - Expression Evaluation & Backtracking
L-11	11	Infix, Postfix, Prefix Expressions - Conversion & Evaluation
L-12	12	Introduction to Queues - Types and Operations
L-13	13	Deque & Priority Queues - Concepts & Applications
L-14	14	Implementation & Applications of Queues and Priority Queues
	Unit III	**Lists**
L-15	15	Introduction to Linked Lists - Singly, Doubly, Circular
L-16	16	Operations on Linked Lists - Traversal, Insertion, Deletion
L-17	17	Searching in Linked Lists & Use of Headers
L-18	18	Applications & Variations of Linked Lists
	Unit IV	**Trees**
L-19	19	Introduction to Trees - Terminology & Types
L-20	20	Binary Tree Traversals - Inorder, Preorder, Postorder
L-21	21	Recursive Algorithms for Tree Operations
L-22	22	Binary Search Tree - Insertion, Deletion, Search
L-23	23	Balanced Trees & Applications
	Unit V	**B-Trees**
L-24	24	Introduction to B-Trees - Concept & Invention
L-25	25	B-Tree Operations - Insertion & Deletion
L-26	26	B-Trees in Database & Filesystems
	Unit VI	**Sorting & Searching Techniques**
L-27	27	Sorting Algorithms - Insertion Sort & Selection Sort
L-28	28	Merge Sort & Heap Sort - Concepts & Implementation
L-29	29	Searching Techniques - Linear Search & Binary Search
L-30	30	Hashing - Concepts, Hash Functions & Collisions

L-31	31	Comparing Sorting Algorithms - Time & Space Complexities
L-32	32	Practical Applications & Problem-Solving Sessions
	33	Mock Tests & Q&A on Entire Syllabus Covered So Far
		Advanced Topics & Revision (Lectures 34-40)
L-33	34	Introduction to Advanced Data Structures (Heaps, Graphs)
L-34	35	Applications of Data Structures in Real-World Scenarios
L-35	36	Algorithm Design & Analysis - Big O, Space & Time Complexities
L-36	37	Case Studies & Problem-Solving Sessions
L-37	38	Revision of Units I to VI
L-38	39	Practice Exam & Doubt Clearing
L-39	40	Final Revision & Tips for Exams
		Assignment
		Activity/ Quiz

1. Chalk & Board
2. PPT
3. Audio-Visual
4. Assignment
5. Models/Activity/ Quiz
Delivery Method
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